



An efficient object detection system for indoor assistance navigation using deep learning techniques

Mouna Afif¹ · Riadh Ayachi¹ · Yahia Said² · Edwige Pissaloux³ · Mohamed Atri⁴

Received: 17 September 2020 / Revised: 31 May 2021 / Accepted: 31 January 2022 /
Published online: 3 March 2022

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2022, corrected publication 2022

Abstract

Building new systems used for indoor objects detection and indoor assistance navigation presents a very crucial task especially in artificial intelligence and computer science fields. The number of blind and visually impaired persons (VIP) is increasing day by day. In order to help this category of persons, we propose to develop a new indoor object-detection system based on deep convolutional neural networks (DCNNs). The proposed system is developed based on the one-stage neural network RetinaNet. In order to train and evaluate the developed system, we propose to build a new indoor objects dataset which also presents 11,000 images containing 24 indoor landmark objects highly valuable for indoor assistance navigation. The proposed dataset provides a high intra and inter-class variation and various challenging conditions which aim to build a robust detection system for blind and visually impaired people (VIP) mobility. Experimental results prove the high detection performances of the developed indoor objects detection and recognition system. We obtained a detection accuracy reaching up to 98.75% mAP and 62 FPS as a detection speed.

Keywords Indoor navigation · Blind and visually impaired persons · Deep learning · Deep convolutional neural networks (DCNN) · Indoor object detection

✉ Mouna Afif
mouna.afif@outlook.fr

¹ Laboratory of Electronics and Microelectronics (EμE), Faculty of Sciences of Monastir, University of Monastir, Monastir, Tunisia

² Electrical Engineering Department, College of Engineering, Northern Border University, Arar, Saudi Arabia

³ LITIS Laboratory & CNRS FR 3638 University of Rouen Normandy, Rouen, France

⁴ College of Computer Science, King Khalid University, Abha, Saudi Arabia

1 Introduction

Indoor object detection and recognition present a very challenging task especially in the computer vision area. Indoor assistance navigation of blind and sighted persons poses a big problem, yet it is worth tackling.

The complexity of indoor assistance navigation lies in the fact of fully understanding the surrounding environment and thus to improve the life quality of blind and sighted persons so they can better integrate and participate in the daily life.

Visual impairments and blindness affect a large number of individuals around the world. Precisely locating and identifying indoor objects and walking confidently, is a very simple task for a person with good sight. However, this task poses a very big problem for the blind and the visually impaired persons. For this fact, it is very important to develop new systems and applications to detect and recognize landmark indoor objects to fully assist blind and sighted persons to explore their surroundings and to more integrate in the daily life. Detecting a specific set of indoor objects presents a very challenging problem as many objects present very little portion from the input image size. Also, other objects are occluded with others which make the detection harder. All these issues are addressed in our work, which makes our proposed application very original compared to other previous works.

In our daily life, a huge amount of visual data is created every split second. For this fact, we need to create new applications and alternatives to help blind, partially sighted and sighted persons as well. The huge growth known in visual information data leads to new challenges faced to process and extraction of the most relevant features which will in turn highly facilitate the image processing process.

According to the latest statistics of the World Health Organization (WHO) [29], 185 million persons suffer from visual impairments around the world. Also, 246 million persons present moderate to severe visual impairments and 39 million are totally blind.

Blind and sighted persons face many navigation problems during their everyday life activities. Finding destinations, searching, walking, detecting objects, avoiding obstacles are difficult tasks for blind and impaired persons. Hence, developing new supporting systems to help this category of persons to navigate safely and to explore their surroundings has become absolutely essential.

Indoor navigation is a very difficult task for sighted, partially sighted and blind persons as well. Indoor object detection and recognition presents a high-valuable artificial intelligence task as it provides the blind and the impaired person by high perceptual abilities which help them to move on and explore their surroundings.

In this paper, we propose to create a new indoor dataset as well as we propose a new deep learning-based system for indoor object detection. By building a reliable indoor object detection system with an affordable cost, we ensure a better life quality for this category of people. We propose in this work a new assistive application which gives the opportunity for the blind or the sighted person to navigate safely in unfamiliar indoor environment by recognizing, detecting and classifying the indoor objects present in the surrounding environment. The proposed indoor object detection and recognition system is robust enough as it was trained using various challenging conditions like various lighting conditions, different objects viewpoints, heavy occlusion, intra and inter-call variation, etc....

Detecting and recognizing indoor objects and estimating their poses in images and videos present a wide range of applications in artificial intelligence and robotics field. Developing high-level object detectors has a fundamental impact on robot manipulation, human robot

interaction, indoor assistance navigation and augmented reality. However, indoor environments always present very complex background, different lighting conditions, heavy occlusion remains the problem as challenging. Furthermore, indoor objects may appear under different shapes, colors, textures and forms depending on the camera acquisition, calibration and viewpoint. Therefore, by developing a new and robust system used for indoor assistance for blind and impaired persons is crucial for daily interaction with real-world indoor environments.

The capability of automatically detecting indoor objects in cluttered indoor environments is a key component of various tasks in computer vision area.

In our approach, our problem remains more challenging especially in indoor environments as they present various lighting conditions, cluttered environments and complex decorations. It will be very useful to develop new indoor object detector to solve the perception problem for both blind and sighted persons. The indoor object detection and recognition task presents a crucial process that is widely used in computer vision and image processing area in order to specify class by giving their locations in images (bounding boxes: bbox) and object classes names. Generally, developing an indoor object detection system require a huge amount of data in order to train and test the deep convolutional neural network and extract high-level features from input images. Basically, an object detection algorithm consists of two main parts: one for classification which provides the object class while the regression part provides the object location (bbox) coordinates.

The aim of the object detection task is to precise whether the input image or video presents a specific class object and estimate the object location by outputting the bounding box which overlaps the object + its class name. This task forms a primary step in various computer vision tasks like object recognition [31], robot navigation [28], road sign detection [24] and image segmentation [25]. By applying and considering the advantages of deep learning algorithms, we propose in this paper to develop a new indoor object detection system based on the famous one-stage deep convolutional neural network named “RetinaNet” [20].

Our aim from this work is to develop a new robust deep learning-based indoor object detection system which will accurately locate and classify a specific set of objects present in the input indoor scenery. The proposed system was trained and tested on our new proposed indoor dataset that was not previously studied. The proposed dataset provides 11,000 indoor images containing 24 indoor landmark objects highly valuable for blind and visually impaired persons (VIP) in indoor assistance navigation. The imported value from our work compared to the state-of-the-art, is that the proposed system was trained and tested under various challenging conditions in order to ensure the robustness of the detection process and also to increase the security of indoor navigation for the blind and the sighted person. The obtained results outperform those of previous works in terms of detection accuracy as well as the detection speed. We obtained 98.75% as a mean average precision of the detection accuracy and 62 frames per second FPS.

The reminder of this paper is the following: section 2 provides the state-of-the-art works related to indoor objects detection field. Section 3 overviews the main contributions of this work. Section 4 details the proposed architecture for indoor object detection and indoor assistance navigation. Section 5 provides the main results and experiments carried out in the proposed work and section 6 concludes the paper.

2 Related works

We are living in a digital world full of digital cameras connected to our computers and smartphones. There is a growing need for intelligent machines to analyze and understand the visual contents. This task will widely help and improve the life quality of blind and impaired persons.

Although, current graphic processor units GPUs and computers can handle extensive computation process, they were widely used to develop and implement new objects detection systems. In order to help blind and sighted persons to better explore their surroundings and participate in the daily life, we have to build a new intelligent system with a higher level of decision.

To be realistic and practical: our proposed detection system addresses the following issues:

- Real world imaging factors: real world images may appear under various lighting conditions. Objects also may appear under different forms, colors, shapes and viewpoints. Also, the objects categories appearance changes significantly depending on the viewing angle.
- Inter-class variation: objects belonging to different classes may have near shapes, colors, forms and materials. For this fact, when building a detection system, we have to pay attention to this issue.
- Intra-class variation: within the same category, objects may appear under different shapes, texture and sizes.
- Background complexity: generally, indoor sceneries presents complex and cluttered environments.

Objects detection and recognition engender a challenging problem especially, when it is applied to cluttered indoor environments which present complex background which makes the object detection task harder.

The major aim of the proposed work is to develop an efficient and robust indoor object detection based on one-stage detectors that would help blind and sighted persons to freely navigate in their surroundings. As the indoor object detection and recognition present a very important task in image processing and artificial intelligence field, numerous works have been proposed in the literature to address this problem.

There have been a large number of works elaborated to develop new systems for object detection used for various applications as: indoor object classification [2, 3], indoor object detection [4], Scene classification [5] and road sign detection [6].

Various academic works have been elaborated aiming to develop new systems for indoor navigation assistance. Simultaneous Localization and Mapping (SLAM) techniques present a very useful alternative to this end. In [14], authors proposed a method based on distributed practical filter simultaneous localization and mapping (DPSLAM) used for indoor positioning. The proposed system presents a suitable technique for persons with no prior-knowledge of the area of operation before the system deployment. Many other works used SLAM techniques for indoor navigation and positioning [10, 32]. There is an increasing need to develop new technologies to enhance indoor navigation and wayfinding for sighted people. To tackle this problem, authors in [18] developed a digital sign system; the prototype system demonstrates its feasibility for enabling visually impaired persons to navigate. Object detection task

can contribute to different other sub-tasks as pedestrian detection [30], face detection [27] and generic object detection [7].

Many classical works on indoor object detection basically rely on classical machine learning techniques [12, 23]. RGB-D sensors provide not only information about the input data, but also depth information. RGB-D cameras are widely employed in indoor assistance navigation works. In [16], authors proposed an object recognition system used for indoor robot navigation using an RGB-D sensor. The system has been developed using random forest classifier and it shows a big effectiveness and a real time suitability for mobile implementations. Blum et al. [9] proposed an object recognition system using RGB-D data. This system has been evaluated using RGB-depth data used for different computer vision tasks. Using depth sensors is very challenging for simultaneous localization and mapping (SLAM) techniques. In [11], authors introduce an object recognition framework for SLAM in indoor environments using depth sensors.

Recently, deep learning -based algorithms have shown big performances as well as impressive results on indoor object detection tasks. Zia et al. proposed in [33] an RGB-D object recognition system based on deep convolutional neural networks. To develop the object recognition system, authors proposed a hybrid 2D/3D convolutional neural network and they obtain very interesting results. In [13], authors developed a pipeline used for indoor object detection based on cafenet neural network. The developed indoor objects recognition system shows good abilities on recognizing a set of landmark indoor objects. Authors in [19] developed an object detection system especially used for detecting small samples using deep learning techniques. In [8], authors proposed to develop an object recognition system using physically- realistic synthetic depth scenes. In [26], sun et al. proposed to address the problem of RGBD-based detection and analysis of waste objects for nuclear decommissioning. This system presents the potential to be applied for other types of applications. This system can also operate on real-time pipeline for RGBD-based data. In [22], liu et al. proposed to develop a tracking control system used to under-actuated based on deep learning techniques. Authors proposed a new adaptive control scheme based on a multi-layer neural network. In [21], authors proposed to study the problem of adaptive trajectory tracking control for a microcapsule system. In their work, authors divided the feasible motion trajectory is derived into dynamic constraints and control indexes.

The proposed indoor object detection system was evaluated using the proposed indoor datasets which provides various challenging conditions. We achieved high detection accuracies as well as encouraging detection speed. In this work, we treated new indoor objects that were not considered by the state-of-the-art datasets as well as we treat dangerous situations to be avoided to ensure safer indoor navigation for blind and impaired persons.

3 Contributions

- The proposed indoor object detection system consists of using a one-stage DCNN model.
- The proposed detection system achieved a very encouraging accuracy for indoor object.
- In this paper, we propose a new indoor object detection dataset consisting of 11,000 images containing 24 landmark indoor objects.
- We provide new indoor object that was not taken into account by other indoor datasets that are highly recommended for blind and sighted people's indoor navigation.

- The proposed system was trained and evaluated in very challenging conditions like extreme lighting conditions, occlusions, high intra and inter-class variation.
- The proposed work was trained and tested using real world images that do not undergo any pre-processing for image quality enhancement.
- Finally, the proposed indoor object detector was trained and tested using the proposed dataset that was not previously studied.

3.1 Dataset collection

The data collection and labeling present a key component to build an accurate and robust object recognition and detection system. Moreover, the annotation of a huge number of data is a labor-intensive problem.

In contrast to the exiting indoor datasets, the proposed dataset provides many challenging conditions taken into account as: heavy occlusions, different lighting conditions, complex background, etc.... in order to increase the robustness of the indoor object detector. Also, the proposed dataset provides a high inter and intra-class variation to build an accurate detector. The following Fig. 1 provides an example of intra and inter-class variation.

The deep variability present in the proposed indoor dataset in term of background complexity, occlusion, blurring, variation of luminosity, dangerous situations to be avoided for the blind and the sighted persons are the main specificity of the proposed dataset.

Indoor scenery presents a huge difference compared to other spaces as it is composed of several objects and decorations (e.g. doors, signs, stairs, elevators, corridors, etc.,...).

The following Fig. 2 presents a wide intra-class variation between doors captured in the proposed indoor dataset. Doors are under different colors, sizes, shapes and textures. Also, doors present in the proposed work are from various materials; some are in wood others in iron and others are in glass. Also, doors are taken under different camera-views and different poses and views as some are opened and others are closed. All these queries make our proposed indoor object dataset suitable to train and test new indoor objects detectors. The proposed indoor dataset is original in term of:

- Light Invariance: this base presents more different images with different light intensity.
- Invariance to rigid geometrical change: the objects in this base are taken from different angles.
- Occlusion (possibility that part or even the entire object is hidden).
- Presence of essential objects for the visually impaired and handicapped.



Fig. 1 Inter-class variation



Fig. 2 Intra-class variation example

- Highlighting of dangerous situations (to be avoided to ensure safe navigation for the visually impaired (down stairs)).
- Presence of images of crossings and intersection of corridors (which give the visually impaired the possibility of decision making).
- Presence of obstacles in the corridors obstructing the passage and presenting safety problems.
- 11,000 images acquired in different locations inside the building in August 2019.
- 24 classes of interior objects.
- This database will be used to train convolution neural networks to develop new detection systems for indoor objects.
- The proposed database presents images of 4032×3024 pixels.
- The originality of this database comes from two new facts taken into account:
- The spatial relations between the objects present in the scene
- The possible actions to be applied to these objects (relations between VIPs and an object).

This set of captured images has many specifications and strong points, as it gathers various data in various lighting conditions and against a complex image background, in order to guarantee greater robustness when detecting these objects via a detection system.

A strong point in our image database, is that it provides a high inter-class variation. That is, classes that are very similar such as normal doors and elevator doors. The following figure shows an example of the intra-class variation.

The database created presents 24 essential and indispensable classes to be recognized for the visually impaired when navigating inside buildings (fire extinguisher, door, ascending or descending stairs, signs, windows,...). The following Table 1 summarizes all the classes of objects present in our database.

The proposed indoor dataset consists of various 11,000 indoor images containing 24 indoor landmark objects essential for daily indoor navigation of blind and visually impaired persons.

Table 1 Classes names

fire extinguisher	trash bin
door	Printer
person	Heater
light switch	Microwave
showcase	Plant
exhibition table	Window
exit	disabled exit
elevator	water dispenser
table	drink dispenser
elevator	Wc
stairs	security button
confidence zone	podotactile tape

The presented dataset presents a fully-labeled data highly recommended to train and test powerful deep learning models to build perform indoor object detection systems. This database is especially proposed to help blind and sighted persons.

for their indoor mobility in indoor environments as: clinics, universities, schools, hospitals, and so on...

The collected dataset will be used to evaluate new indoor object detection system built for blind and sighted person's assistance navigation. The proposed dataset provides a multi-level of object rightness and it present a step forward in new applications to help a huge category of people.

3.2 Proposed architecture for indoor object detection

The visual system of human being is very complex and powerful as it consists of various components. It is efficient enough and extremely fast to detect and recognize various objects at the same time. Despite the complexity of our perception system, it is extremely hard to develop an object detection system.

To tackle this problem, we propose to develop a new indoor object detection system used for blind and sighted people in indoor assistance navigation to independently navigate and access new unfamiliar indoor sceneries.

In order to develop our new proposed indoor objects detection system, we take into account the following issues:

- Various unsafe situations cases to avoid (downstairs, upstairs, etc....).
- Train and test the model under various challenging conditions.
- New indoor object classes to be detected.
- Train and test the model using 11,000 images containing 24 landmark indoor object classes.
- The proposed dataset contains essential objects for blind and visually impaired persons.

In order to give the maximum of information to the blind and the sighted persons, our proposed approach demonstrated a good ability to detect vital indoor objects which were considered for the first time.

Due to the huge development of the deep learning technique in the last few years, it makes unsolvable tasks possible. Blindness and visual handicaps cause a considerable isolation from

the society. To solve this problem, we propose our alternative application to help this category of persons to better participate and integrate in the daily life.

Our proposed work is based on deep convolutional neural network RetinaNet [20]. Generally, neural networks suffer from the problem of class imbalance. Based on this fact, we developed our proposed object detection system based on the one-stage network RetinaNet [20] which proposed new loss function which contributes for better solutions on the class imbalance problem.

RetinaNet architecture is based on two main parts:

- A feature extractor built on the top of the ResNet 50 [15] extractor called feature pyramid network FPN used to calculate the feature map of the input images or videos.
- Classification head: perform the classification using the backbone's output.
- Regression head: responsible for outputting the bounding box (bbox) coordinates using the backbone's output.

The construction of feature pyramid network (FPN) network consists of two pathways connected with lateral connections. Bottom-up pathway presents a convolutional neural network used for feature extraction. It chooses the latest feature map of the output features map of the same scales. The obtained features will be used for the FPN foundation. FPN network able to construct higher resolution layers despite the constructed layers are semantically strong, but after applying up and down sampling, the objects locations are not perfectly precised. To tackle this problem, lateral connections are added to better detect objects locations.

An input image from the proposed dataset is fed into the network, after that a feature extraction process is performed via ResNet 50 network in order to extract the main features from the input image. After that the extracted features are passed through the FPN network in order to calculate features maps and to perform up-sampling and down-sampling operation in order to maintain only the images region that may contain objects. Then, all the features map are fed through the classification and regression head to extract the object's classes and places in the images.

To improve the detection performance, RetinaNet provides a simple but very effective new loss function called “focal loss” which enables us to achieve better results on difficult samples. We note that RetinaNet achieves the state-of-the-art in term of detection accuracy and running time.

RetinaNet architecture is much simpler than two-stage-based architectures which are composed of independent networks for classification and regression. Focal loss is proposed to address the problem of class imbalance between foreground and background during the training stage of the one-stage object detector. Focal loss is the effective extension of cross-entropy function.

By employing the cross entropy during the training process generates successful results when the model is trained on a balanced set. However, it is unsuitable for one-stage detectors. Basically, for one stage detectors when calculating the loss over actual objects and the generated samples, the loss of the easy common samples of background can be much greater than the loss of difficult uncommon samples. To address this important problem, the focal loss function introduces a modulating factor $\alpha_t(1 - p_t)^\gamma$ applied to the Cross entropy (CE) function as mentioned in Eq. 1.

$$FL(P_t) = -\alpha_t(1-p_t)^\gamma \log(P_t) \quad (1)$$

While $\alpha_t \in [0,1]$ for class 1 and $1-\alpha_t$ for class -1 . For national convinces they define P_t . The balanced cross entropy function can be defined as, the following Eq. 2.

$$CE(P_t) = -\alpha_t \log(P_t) \quad (2)$$

$$P_t = \begin{cases} 1 & \text{if } y = 1 \\ 1-P, & \text{otherwise} \end{cases} \quad (3)$$

Focal loss function focuses on down-weighting the easy samples and focusing more on training the neural network on the hard examples. The focal loss function presents two main aspects:

- For a small value of P_t , the modulating value is near 1 and the loss function remains unaffected. For $P_t \rightarrow 1$, the modulating value goes to 0 and the loss for the well-classified samples is down-weighted.
- For the focusing parameter γ , it adjusts smoothly the down-weighted of the easy sample rate. If $\gamma = 0$, FL is equivalent to CE. If γ presents a high value, the effect of the modulating parameter is increased and based on this fact, the focal loss leads to improving the network accuracy.

By introducing the modulating value, one-stage object networks can present various forms of loss functions which more concentrate more on training the network on difficult samples.

For binary classification problems, models are initialized to have equal outputs ($y = -1, y = 1$). Based on model's initialization and the presence of class imbalance, the most frequent classes' loss can dominate the total loss value. To address this problem, RetinaNet introduces a prior-value of p-estimated used for the rare classes (foreground) during the training step. This technique highly stabilizes the training process.

Region proposal network (RPN) performs features map sub-sampling which crosses a lot of important image features which contributes to a lack on the small object detection. To address this problem, FPN are introduced to extract important features maps from the entire input images. FPN backbone is built on the top of ResNet architecture and containing two subnets: one for classification and the second for classification.

The proposed indoor object detection system is based on RetinaNet with the use of ResNet 50 as a network backbone. The RetinaNet-based architecture for object detection is presented in Fig. 3.

FPN network performs features-map extraction at different scales. By using the FPN network, the multi-scale object detection is highly improved. Detecting objects with different sizes (small, medium, and large objects) poses a serious problem especially when detecting small objects. For each scale stage, one feature pyramid layer is associated. Each last output is used as feature map to feed the top-down pathway via lateral connections.

After all this work, a huge number of anchor boxes is detected, to reduce this number, a non-maximum suppression is applied to select only boxes with the highest confidence scores. Eventually, all the remaining anchors will be considered as bounding boxes.

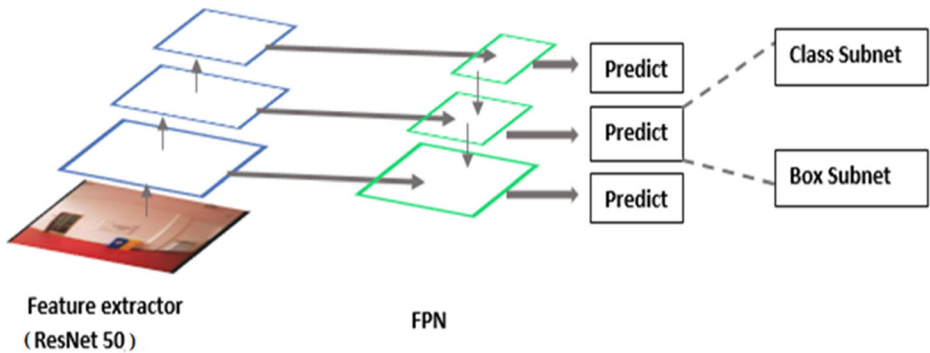


Fig. 3 RetinaNet-based architecture used for indoor object detection

4 Experiments and results

Indoor scene assistance navigation for blind and sighted persons requires a real-time processing and high-performance accuracy.

In this paper, we take a further step to tackle the indoor object detection problem. Our experiments were performed on the proposed indoor dataset. Images provided in the proposed dataset were taken in real indoor environments.

4.1 Training experiments

This dataset consists of 11,000 images containing 24 vital landmark indoor objects. Our collected dataset presents various complex conditions like complex illumination conditions, various points of view of objects, heavy occlusions and so on.

For the experiments, we split the dataset into two parts: one for training and the second for testing the DCNN model. During the proposed experiments, we used 7300 images to train the model and 3700 for testing. Parameters' settings used during training process are the following: we set the learning rate of the ADAM optimizer [17] to 0.0001, we trained the DCNN models for 50 epochs each contain 10,000 iterations and we fixed the batch normalization size to 100. The images provided in the proposed dataset are of size 4032×3024 pixels. The proposed experiments were run on a HP workstation equipped with an intel Xeon E5-2683 v4 processor, a tesla k40c graphic processor of 12 G of graphic memory. The implementation of the overall work was performed using Keras framework which present a high-level neural network programming interface. To develop the proposed indoor object detection system code, we used python 3.6 setup, TensorFlow [1] 1.13 NVIDIA CUDA toolkit 10.0 and the deep learning library CUDNN 7.0. As an evaluation metric used for the proposed indoor object detection system we used the mean average precision (mAP).

In this paper, we address the problem of blind and impaired individual's indoor navigation to avoid dangers and obstacles and safely reach their desired destinations. We evaluate and test the proposed detection system on our proposed multi-object dataset. We proposed this dataset to cover very challenging conditions and dangerous situations to be avoided when blind and sighted person is navigating. Based on the obtained results, we believe that the proposed indoor object detection system highly contributes to a safer way of indoor navigation for blind and impaired persons.

The proposed system for indoor object detection and blind's indoor assistance navigation was evaluated using the proposed indoor dataset. We concentrated on 24 landmark indoor objects highly recommended for indoor navigation which were not proposed by state-of-the-art datasets. These proposed objects are highly relevant to improve the life quality for blind and sighted persons. The proposed architecture was developed by using python 3.6 and keras framework.

In order to highly rely on the indoor assistance navigation for blind and sighted persons, we have to develop a robust detection system which can operate in real time conditions.

The proposed work takes a step further to improve the life quality of a huge number of persons around the world by applying the techniques of deep learning and deep convolutional neural networks.

The average precision of every indoor object class present in the proposed dataset, is the evaluation criteria for our proposed approach.

To train a DCNN, we need a huge amount of data. For this fact, we propose the new indoor dataset "LITIS data" to feed the neural network. To ensure more robustness and efficiency for the proposed work, we trained the neural network on various challenging conditions. Images provided in the LITIS data present various objects in the same image which makes them suitable for detection issues. We provide in the proposed indoor objects dataset new landmark objects that were not mentioned in the state-of-the-art datasets and which are vital for indoor navigation of blind and sighted persons. The following Table 2 provides all the training settings used when carrying out our proposed experiments.

The dataset was divided into two main parts: one for training and the second for testing. During our experiments, we used 7300 images for training and 3700 images for testing. Training a deep convolutional neural network requires a huge amount of annotated data. For this fact, we applied data augmentation to increase the number of training data and to avoid the risk of the network overfitting. Data augmentation was applied by using image rotations, flipping and translations.

As a network optimizer, we used ADAM which updates the network parameters quickly. The biggest strength of using ADAM as a focal loss function optimizer is that it computes an adaptive learning rate value for each network parameter during the training step.

4.2 Testing experiments

Testing experiments were performed on the same desktop of the training step. To evaluate the performance of the proposed indoor object detection system, during the training process, we included the input images with their corresponding ground truth bounding boxes (bbox) which

Table 2 Experiments settings

epochs	50
steps	10,000
Learning rate	1×10^{-5}
Min-image rescale	800
Max-image rescale	1333
Optimizer	ADAM
Batch size	1
Regression loss	Smooth-l1
Classification loss	Focal loss

contain the positions of objects. While testing the network, the detected ground truth bounding boxes are compared to the overlapping regions between the detected boxes and the ground truth boxes. The next Table 3 provides the average precision per-class obtained during the proposed experiments.

As mentioned in Table 3, we obtained very encouraging results coming up to 100% as a detection precision for various indoor class objects as (window, showcase, exhibition table, table, confidence zone, security button, trash bin, water dispenser, drink dispenser, sofa, microwave, plant, printer, disabled exit and wc). We also note that we obtained 98.75% as a mean average precision (mAP) for the whole dataset.

In order to obtain more information about the proposed indoor objects detection system performances, we calculated various valuations metrics including precision, recall and F1-score, true positive (TP), false positive (FP) and false negative (FN). All these metrics are calculated as the following equations. Table 4 provides all the obtained results for the different evaluations metrics used during the proposed experiments.

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

$$F1-score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (6)$$

Table 5 present a per-class comparison between the obtained per-class average precision and those obtained in our previous work [4]. We note that we obtained the high per-class average precision compared to our previous work. We also note that the proposed dataset presents the first indoor object dataset presenting 24 indoor landmark objects highly recommended for indoor navigation. To benchmark and evaluate the accuracy of our model, we used the mean average precision (mAP) metric. Detection accuracies comparison per-class is mentioned in Table 5.

Our proposed work is especially dedicated for blind and sighted persons to help them to better explore their surroundings. As far as we consider the visually impaired person's mobility, the proposed indoor object detection system should provide an ahead inference about the upcoming objects before the sighted person reach the desired object. We can assume that:

- The optimal distance between the desired indoor object and the VIP, which is sufficient to detect and recognize it in advance, is about 5 m.
- We consider that the speed of the VIP is about the speed of a normal person (1.4 m/s).

By considering all the above, the VIP person needs 3.57 s to reach a desired object situated from 5 m away from him. Our proposed system achieves a good time processing of 62 frames per second (FPS) which meets the requirements of the sighted persons and the real time constraints.

Table 3 Results obtained when using ResNet 50 as a feature extractor

Class name	Window	door	Person	Light switch	Showcase	Exhibition table	exit	stairs
AP (%)	100	83.60	98.64	99.83	100	100	100	99.97
Class name	table		Security button	Podotactile tape	Trash bin	Water dispenser	heater	sofa
AP (%)	100	100	100	89.46	100	100	99.33	100
Class name	microwave	plant	Fire extinguisher	Printer	Disabled exit	wc	elevator	Drink dispenser
AP (%)	100	99.99	99.83	100	100	100	99.47	100

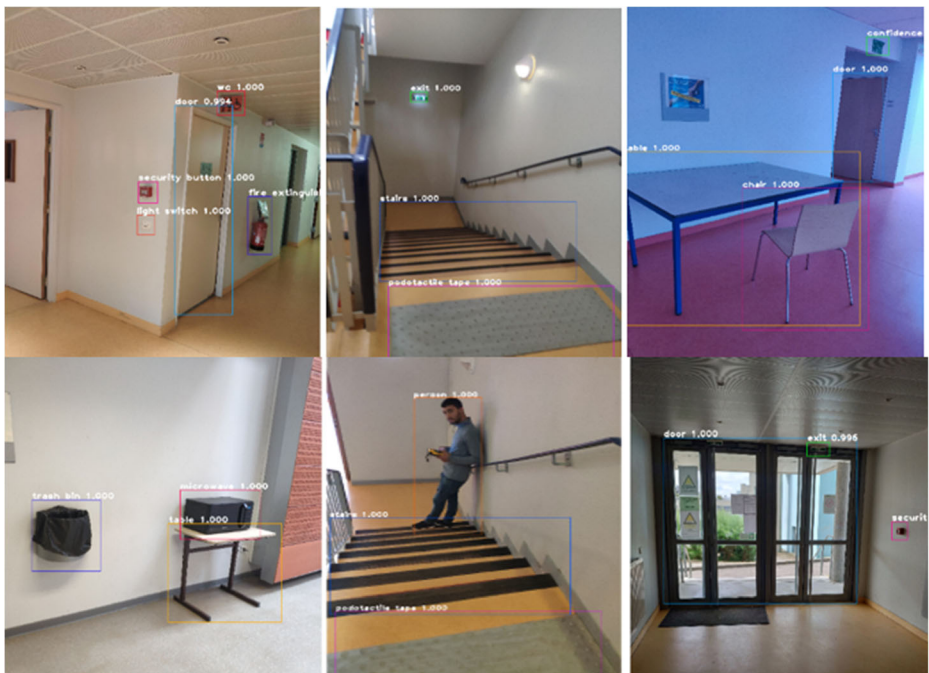
Table 4 Evaluations metrics used

Precision	95.02
Accuracy	98.75
Recall	96.5
F1-score	95.75
True Positive (TP)	11,920
False Positive (FP)	624
False Negative (FN)	432

Table 5 Per-class comparison results

Class name	window	elevator	door	Light switch	Trash bin	stairs	Security button	heater	Fire extinguisher	table
Previous work [4]	69.35	95.43	98.97	54.11	94.39	90.65	67.73	98.13	85.34	98.93
Proposed	100	99.47	83.60	99.83	100	99.97	100	99.33	99.83	100

Figure 4 provides an example of detection, as presented all the objects present in the input images are detected with high detection accuracies. We note that the objects are detected whatever their sizes, colors, shapes and luminosity intensities.

**Fig. 4** Detection results illustration

We can conclude that our proposed indoor object detection system has shown the best results in terms of detection accuracies as well as the number of indoor objects treated compared to the state-of-the-art works.

As mentioned in Fig. 4, we note that all the objects presented in the images are successfully detected with high detection rates. The main advantage from proposing this work is that it is highly recommended for blind and visually impaired person's indoor assistance navigation. The proposed indoor objects detection and recognition system achieves very competitive results in terms of detection accuracy as well as processing time. We obtained high detection performances which outperforms those obtained by the state-of-the-art works. Despite the huge performances obtained, the developed system cannot be deployed on embedded system under its current size as it is composed of huge number of parameters. In order to address this problem, we propose as a future work to decrease the network size to fit the capabilities of low-end devices by using quantization and pruning technique.

5 Conclusion

In this paper, we propose a very efficient indoor object detection system based on a one-stage neural network. The proposed detection system was trained and tested using the new proposed indoor dataset which covers various challenging cases and provides new landmark indoor objects that was not presented in the previous works. The proposed indoor object system enables the blind and the visually impaired person (VIP) as well to recognize a set of 24 indoor landmark object highly relevant for its indoor navigation in order to be familiar with its surroundings. The proposed dataset contains 11,000 indoor images containing 24 indoor objects classes. Images presented in the proposed dataset present various challenging conditions in order to build new robust detection system. The proposed dataset provides various images and situations highly relevant for the blind and sighted people's mobility. We achieved the highest detection accuracy, which comes up to 98.75% and 63 FPS as a detection running time. The obtained results outperform those of the state-of-the-art.

Acknowledgements The authors extend their appreciation to the Deputyship for Research & Innovation, Ministry of Education in Saudi Arabia for funding this research work through the project number “IF_2020_NBU_206”.

References

1. Abadi M, Barham P, Chen J, et al. (2016) Tensorflow: A system for large-scale machine learning. In : 12th {USENIX} Symposium on Operating Systems Design and Implementation ({OSDI} 16). 265–283.
2. Afif M, Ayachi R, Said Y et al. (2018) Indoor image recognition and classification via deep convolutional neural network. In : International conference on the Sciences of Electronics. Technol Inform Telecomm. Springer, Cham p. 364–371.
3. Afif M, Ayachi R, Said Y et al. (2019) Indoor object classification for autonomous navigation assistance based on deep CNN model. In : 2019 IEEE International Symposium on Measurements & Networking (M&N). IEEE. 1–4.
4. Afif M, Ayachi R, Said Y et al. An evaluation of retinanet on indoor object detection for blind and visually impaired persons assistance navigation. Neural Process Lett, 2020, p. 1–15

5. Afif M, Ayachi R, Said Y et al (2020) Deep learning based application for indoor scene recognition. *Neural Process Lett* 51:1–11
6. Ayachi R, Afif M, Said Y et al. Traffic signs detection for real-world application of an advanced driving assisting system using deep learning. *Neural Process Lett*, 2020, vol. 51, no 1, p. 837–851.
7. Bai Z, Kasun LLC, et Huang G-B. Generic object recognition with local receptive fields based extreme learning machine 2015, 53, 391, 399.
8. Baimukashev D, Zhililbayev A, Kuzdeuov A, et al. Deep learning based object recognition using physically-realistic synthetic depth scenes. *Mach Learn Knowledge Extraction*, 2019 1(3): 883–903.
9. Blum M, Springenberg JT, Wülfing J, Riedmiller M(2012) A learned feature descriptor for object recognition in RGB-D data," in *Robotics and Automation (ICRA)*. IEEE Int Conf, 1298–1303.
10. Buyval A, Afanasyev I et Magid E (2017) Comparative analysis of ROS-based monocular SLAM methods for indoor navigation. In : *Ninth International Conference on Machine Vision (ICMV 2016)*. Int Soc Optics Photonics. 103411K.
11. Chae HW, Park C, Yu H, Song JB (2016) "Object recognition for SLAM in floor environments using a depth sensor," in *Ubiquitous Robots and Ambient Intelligence (URAI)*. 2016 13th Int Conf, 405–410
12. Ding W, Gu JN, Tang SX, Shang ZY, Duo EA, Zheng CJ (2016) Development of a calibrating algorithm for Delta Robot's visual positioning based on artificial neural network. *Optik-Int J Light Electron Optics* 127(20):9095–9104
13. Ding X, Luo Y, Yu Q et al. (2017) Indoor object recognition using pre-trained convolutional neural network. In : *2017 23rd International Conference on Automation and Computing (ICAC)*. IEEE. 1–6.
14. Faragher RM, Samo C et Newman M (2012) Opportunistic radio SLAM for indoor navigation using smartphone sensors. In : *Proceedings of the IEEE/ION Position, Location and Navigation Symposium*. IEEE, 2012. 120–128.
15. He K, Zhang X, Ren S et al. (2016) Deep residual learning for image recognition. In : *Proceedings of the IEEE conference on computer vision and pattern recognition*. 770–778.
16. Jiang LX, Koch A, Zell A (2016) "Object recognition and tracking for indoor robots using an RGB-D sensor," in *Intelligent Autonomous Systems 13: Proceed 13th Int Conf IAS-13*. 859–871.
17. Kingma DP ,et Ba J Adam: A method for stochastic optimization. arXiv preprint arXiv:1412.6980, 2014.
18. Legge GE, Beckmann PJ, Tjan BS et al (2013) Indoor navigation by people with visual impairment using a digital sign system. *PloS one* 8(10):e76783
19. Li C, Zhang Y, Qu Y (2018) Object detection based on deep learning of small samples. In: *2018 tenth international conference on advanced computational intelligence (ICACI)*. IEEE, pp 449–454.
20. Lin T-Y, Goyal P, Girshick R et al. (2017) Focal loss for dense object detection. In : *Proceedings of the IEEE international conference on computer vision*. . p. 2980–2988.
21. Liu P, Yu H, Cang S (2018) Optimized adaptive tracking control for an underactuated vibro-driven capsule system. *Nonlinear Dyn* 94:1803–1817. <https://doi.org/10.1007/s11071-018-4458-9>
22. Liu P, Yu H, Cang S (2019) Adaptive neural network tracking control for underactuated systems with matched and mismatched disturbances. *Nonlinear Dyn* 98:1447–1464. <https://doi.org/10.1007/s11071-019-05170-8>
23. Mei S, Yang H, Yin ZP (2017) Discriminative feature representation for image classification via multimodal multitask deep neural networks. *J Electron Imaging* 26(1):013023–013023
24. Saadna Y, et Behloul A (2017) An overview of traffic sign detection and classification methods. *Int J Multimedia Inform Retrieval*. 6 (3): 193–210.
25. Satyawana S (2016) A review paper on image segmentation and object recognition procedures. *Sci Eng Technol* 67
26. Sun L, Zhao C, Yan Z, Liu P, Duckett T and Stolkin R (2019) A Novel Weakly-Supervised Approach for RGB-D-Based Nuclear Waste Object Detection. *IEEE Sensors J*. 19(9): 3487–3500, <https://doi.org/10.1109/JSEN.2018.2888815>.
27. Sung KK, Poggio T (2002) Example-based learning for view-based human face detection. *IEEE Trans Pattern Anal Mach Intell* 20(1):39–51
28. Wang C, Meng L, She S et al.(2017). Autonomous mobile robot navigation in uneven and unstructured indoor environments. *2017 IEEE/RSJ Int Conf Intell Robots Syst (IROS)*. IEEE. 109–116.
29. Who: Vision impairment and blindness. <http://www.who.int/mediacentre/factsheets/fs282/en/>. Accessed 8 Jan (2020)
30. Wojek C, Dollar P, Schiele B, Perona P (2012) Pedestrian detection: an evaluation of the state of the art. *IEEE Trans Pattern Anal Mach Intell* 34(4):743
31. Zhang M, Tseng C, et Kreiman G (2020) Putting visual object recognition in context. In : *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. p. 12985–12994.

32. Zhang X, Li BJ, Samleo L, et al. (2015) A slam based semantic indoor navigation system for visually impaired users. In: 2015 *IEEE International Conference on Systems, Man, and Cybernetics*. IEEE. 1458–1463.
33. Zia S, Yuksel B, Yuret D et al. (2017) RGB-D object recognition using deep convolutional neural networks. *Proceed IEEE Int Conf Comput Vision Workshops*. 896–903.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.